scOPAlign · SINGLE-CELL OPERATOR ALIGNMENT · WORK IN PROGRESS

scOPAlign: aligning single-cell batches by their dynamics

Testing whether batch correction at the level of cell-state operators preserves biology

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[QR code to project repository]

01 The problem

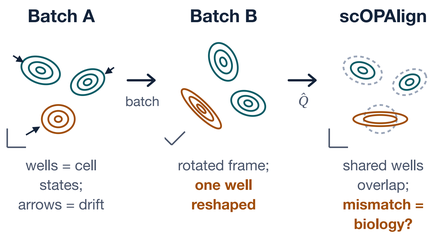
Single-cell experiments run in batches (days, kits, chemistries). A batch shifts every cell's measurements, so one cell type lands in different places.

Common tools (Harmony, scVI) move cells until batches overlap. Because they cannot tell a batch shift from a biological one, they can also erase real biology.

02 Hypothesis

Correcting batches at the operator level, then passing that correction to the cells, preserves biology better than moving cells directly.

Each cell state sits in a "well" of a learned landscape. The *operator* describes the well's shape.



A technical batch re-draws the same wells in a rotated frame. scOPAlign finds the rotation $\hat{Q}$ from the operators and applies it to the cells. A well that biology reshaped cannot be matched by a rotation, so it is kept rather than forced together.

03 The model

1 · DRIFT ON AN ENERGY LANDSCAPE (scJDO2)

$$u(z) = -\nabla E(z)$$

Cells at position z drift downhill on a landscape E learned from the data.

2 · THE OPERATOR

$$J(z) = -\nabla^2 E(z)$$

The curvature of the well around each cell: its shape and stiffness.

3 · A TECHNICAL BATCH IS A ROTATION

$$z_B = Q\, z_A,\ Q^T Q = I$$

$$\Rightarrow J_B(Qz) = Q\, J_A(z)\, Q^T$$

If only the frame changed, the operators match after rotating.

4 · ESTIMATE Q FROM OPERATORS ONLY

$$\hat{Q} = \arg\min_Q \Sigma_i \, \| u_B(Qz_i) - Q\, u_A(z_i) \|^2 + \| J_B(Qz_i) - Q\, J_A(z_i) Q^T \|^2$$

No cell pairing and no labels. Then every cell moves: $z \mapsto Qz$.

5 · WHAT IS LEFT OVER

$$\text{frame ratio} = \| J_B - \hat{Q} J_A \hat{Q}^T \| \div \text{refit noise}$$

≈ 1: as similar as refitting the same data. Clearly above 1 in one well: candidate biology.

04 Data & design

- Public 10x Genomics PBMC data, **one donor**: purely technical batches (v2 vs v3 chemistry; random halves of one run).
- Coordinates: 30-d PCA. Cell types: CellTypist.
- Every test **pre-registered**: thresholds fixed before running.

Mixing: share of neighbours from the other batch (1 = fully mixed).
Silhouette: how well cell types stay apart.

05 Result 1 · Operators are reproducible

Refitting, or fitting on matched cells from another batch, changes operators by about as much as refitting noise (ratio ≈ 0.8–1.5).

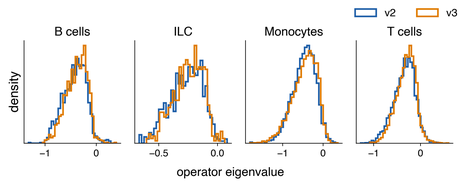


Fig. 1 · Operator eigenvalues in v2 and v3 cells, by cell type (K2).

06 Result 2 · Aligned batches share operators

Rotating v2 into v3's frame with the reference rotation (from gene loadings), operators agree within refit noise: frame ratio **1.09** (repeat: 1.20). Mixing rose **0.11 → 0.90**.

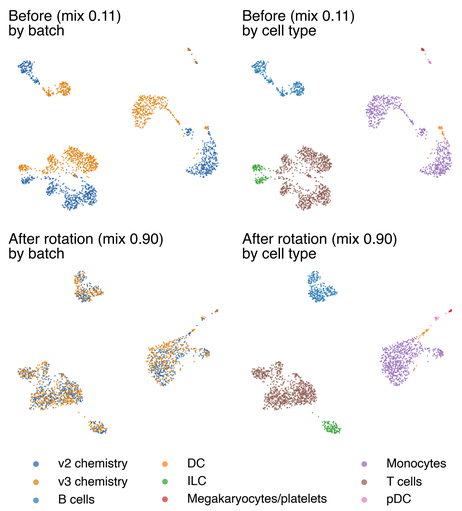


Fig. 2 · UMAP before and after the reference rotation, by batch and by cell type (T02c).

07 Result 3 · Alignment from operators alone

The core of scOPAlign: estimate $\hat{Q}$ from drift and operators only (equation 4).

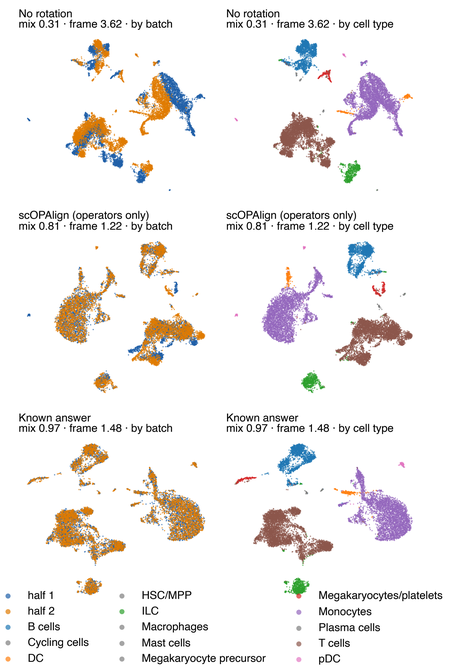


Fig. 3 · No rotation · scOPAlign estimate · known answer (O1b).

- Cells align: mixing **0.81**, frame ratio **1.22**.
- The exact known rotation is not yet recovered (error ≈ 0.6; target ≤ 0.10).
- Curvature alone finds the axes but not their signs; adding drift fixes the signs.
- Operators fix the high-variance directions well and low-variance ones poorly. Next fix: restrict Q to the informative directions.

08 How it compares

Method	Mix ↑	Sil ↑	Frame ↓
Harmony	1.00	0.447	1.35
Reference rotation*	0.90	0.421	1.55

On this purely technical pair Harmony mixes better. *The gene-loading rotation of Result 2, which scOPAlign aims to recover from operators alone (Result 3). The real test needs biology to preserve (10).

09 Road to the test

- ✓ **Operators are reproducible** — Done
- ✓ **Aligned batches share operators** — Done, one technical pair
- ◐ **Rotation from operators alone** — Partial: aligns cells, not yet exact
- ○ **Preserves biology better** — Not yet tested: the hypothesis itself

10 Next: the real test

A simulation where we **know** the batch effect and the biology: one cell state is reshaped in one batch only. Which method keeps that difference: Harmony, scVI, gene-level rotation, or scOPAlign?

11 Conclusions

- Cell-state operators are stable enough to align batches with.
- Operator-only alignment works for cells but is not yet exact.
- Limits: one donor, technical batches, 30-d PCA. No claim yet that biology is preserved better.

Data: 10x Genomics public PBMC datasets · Drift model: scJDO2 · Cell types: CellTypist · Pre-registered analyses; code: [REPO LINK]

[ACKNOWLEDGEMENTS / FUNDING]